

COURSE 3 - DATA ENGINEERING 2026

C3-074B ColdStream Protected Review

Open only after the matching Gate attempt/evidence is frozen.

Artifact	C3-074B-REVIEW
Status	2026 - LEARNER EDITION
Phase	C3-06 Streaming & Event-Driven Pipelines 13 lessons 34-46 focused hours
Prerequisite	C3-05 Gate PASS; C2B streaming/lakehouse concepts may be recalled but Kafka/Spark streaming implementation is taught here.
Current lane	Apache Kafka 4.3.1 Spark Structured Streaming 4.2.0 Delta Lake 4.4.0 Debezium 3.6.2.Final
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: mental picture -> tiny event example -> guided Kafka/streaming action -> expected result -> why it worked -> break/fix -> independent changed case -> validation -> replay/state/production reasoning -> evidence. A running stream is not mastery if duplicates, late data, offsets, checkpoints or sink state are wrong.

Reviewer controls and pass evidence

- Source rows: 20; unique event_id: 19; duplicate rows: 1.
- Unique sensor_id: 5; temperature_c source sum: 93.0.
- Event-time range: 2026-12-03T10:03:00Z -> 2026-12-03T10:54:00Z.
- A passing solution preserves raw source evidence, explicitly handles the duplicate and late event, and never hides them through silent filtering.
- Kafka/Spark/Delta claims are supported by actual runtime evidence when claimed; simulator/design evidence is labeled separately.
- Checkpoint/replay and sink behavior converge to a correct state after failure/restart; no destructive checkpoint editing is accepted.
- Every one of the eight required competencies must be demonstrated; unresolved correctness, integrity, replay, evidence or recovery failure blocks PASS.

Changed recovery

If one competency failed, assign targeted remediation and a changed micro-case. Gate B/C does not auto-open from a failed Gate or from completing repair; a fresh full retest requires explicit assignment.